14 Prove algebraically that the recurring decimal $0.4\dot{5}\dot{7}$ can be written as $\frac{151}{330}$

Let $x = 0.4575757...$

1 mark
Identify repeating57

$10x = 4.575757...$

$1000x = 457.575757...$

Subtract: $990x = 453.$

1 mark
Aligned recurring tails

$$x = \frac{453}{990} = \frac{151}{330}$$

1 mark
Simplify by3

(Total for Question 14 is 3 marks)